qm-notes

kayla

Week 2

Definitions

and

The probability of A and B happening, also the joint probability of $A \cap B$, is computed using the following general rule:

$$\Pr(A \cap B) = \Pr(A|B) \times \Pr(B) = \Pr(B|A) \times \Pr(A)$$

or

The probability of A or B happening (which includes all of *only* A, *only* B, and *both* A *and* B) is computed using the following general rule:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

where we subtract any 'overlap' between A and B which was double counted in adding both.

Independence

An event A is *independent* of another event B iff the probability of A happening is not conditional on the probability of B happening. Formally, the conditional probability of A on B is equal to the marginal probability of A, and therefore equal to the probability of A on not-B. That is,

$$\Pr(A|B) = \Pr(A| \sim B) = \Pr(A)$$

When this happens, the joint probability of A and B is

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B)$$

Mutual exclusivity

An event C is *mutually exclusive* with another event D iff neither event can occur simultaneously. Formally,

$$\Pr(A \cap B) = 0 \therefore \Pr(A \cup B) = \Pr(A) + \Pr(B)$$

This also means that the conditional probabilities of both events on each other $\Pr(A|B) = \Pr(B|A) = 0$.

Collective exhaustivity

A set of events, $E_i = \{E_1, E_2, ..., E_n\}$, is collectively exhaustive when they add up to 1. Formally,

$$\Pr(\bigcup_{i=1}^{n} E_i) = \sum_{i=1}^{n} \Pr(E_i) = 1$$

Classical probability

Classical probability is the study of probability in the absence of data. Therefore, definitions and concept in this paradigm are built upon relations between discrete events and cardinalities of sets. In the absence of data, we conceptualise probability as the ratio between the *cardinality of the set of outcomes in an event* and that of *the sample space*, where the sample space is the set of all possible outcomes.

Letting $E = \{e_1, e_2, ..., e_m\}$ be some event with m outcomes and $S = \{e_1, ..., e_m, e_{m+1}, ..., e_n\}$ represent the sample space,

$$\Pr(E) = \frac{|E|}{|S|}$$

An illustrative example is the theoretical 'concept' of rolling an equal-probability die, let's say a cubical die. In practice, the running *empirical* probability of rolling a die will almost never be perfectly $\frac{1}{6}$ for each side. However—especially relevant given the presumption of equal probability—we can theoretically expect each outcome to have a probability of $\frac{1}{6}$, i.e., the ratio between the cardinalities of the *single-outcome* event* $E \in S$ (1) and the sample space $S = \{1, 2, 3, 4, 5, 6\}$ (6).

Simple and compound events

Simple events—like the *single-outcome event referenced above—are, well, only 'made up' of a single outcome. Formally, simple events are events whose set of outcomes is a singleton ($|E| = 1$). Outcomes themselves can be seen as events if we're interested in examining the outcomes in and of themselves; the outcome–event distinction would only be a conceptual one in such a case.

Compound events, on the other hand, can be made up of multiple outcomes. To return to the die roll example, a compound event would look something like 'rolling an odd number' or 'rolling a number smaller than three'.

From specific to general

Variables in political science are usually stochastic, where whether an outcome (an event happening/taking on a certain value) is true is 'up to chance' and its associated causal mechanisms *change the probability* of it happening.

From a frequentist perspective at least, causal-statistical accounts of political phenomena can thus be viewed as 'repeated trials' on the population, where the probability of an effect (some outcome) manifests as the number of units/observations for which this is true. Observing a causal phenomenon increasing/decreasing this number, then, can be seen as some cause resulting in a higher probability of effect being true.

Random variables

Let Y be some random variable on a set of outcomes O. As Y is a function,

$$Y : O \mapsto \{y_1, y_2, ...\}$$

where $\{y_1, y_2, ...\}$ is the **distribution** (also sample space). Within this distribution, n_1 and n_2 are **realisations** of the random variable. In addition, the **support** of this distribution is the subset of the distribution s.t. $\forall y \in$ support, $\Pr(y) > 0$. Lastly, the function which maps all the members of the distribution to their associated probabilities,

$$f : \{y_1, y_2, ...\} \mapsto \{\Pr(y_1), \Pr(y_2), ...\}$$

is the **distribution *function***.

$\Pr(A) = \sum_{i=1}^{n} \Pr(A|B_i) \times \Pr(B_i)$

Discrete probability distributions

Bernoulli distribution

Bernoulli distributions exclusively concern *binary variables*, describing the frequency of both potential outcomes over repeated observations. It can be represented as follows:

$$\Pr(Y = y|p) = \begin{cases} (1-p) & \text{for} \quad y = 0, \\ p & \text{for} \quad y = 1 \end{cases}$$

In the general form,

$$\Pr(Y = y|p) = p^y(1-p)^{1-y}$$

The Bernoulli distribution is based on the assumption that the trials are independent, which then needs to be justified/self-evident when applying it in analysis.

Binomial distribution

The binomial distribution broadens the size (number of trials) of the Bernoulli distribution$\Pr(Y = y|n, p) = \binom{n}{y} p^y(1-p)^{1-y} = \frac{n!}{y!(n-y)!} p^y(1-p)^{1-y}$

Poisson distribution

$\Pr(Y = y|\mu) = \frac{\mu^y}{y! \times e^{\mu}}$, where $\mu > 0$ is the expected value of the number of events and the variance $\sigma^2 = \mu$.

This PMF describe the number of times one observes 0 events, 1 event, etc. throughout a fixed period of time. Events that can be described by a Poisson distribution are

- discrete (integer count)
- independent (just like Bernoulli), and
- distributed with a *known mean.*

Negative binomial distribution

$\Pr(Y = y|r, p) = \binom{y+r-1}{y} p^y(1-p)^r$, where y is the number of successes (events), r is the number of failures (non-events), $y + r$ is the number of opportunities (or Bernoulli trials), and p is the probability of a success. Events that can be described by a negative binomial distribution

- are independent
- have two possible outcomes (binary)
- keep a constant probability of successes from one trial to the next
- continue until a total of r successes (/failures) are observed, where r is fixed

Negative binomial distributions allow applications to decouple the variance from the mean.

Continuous probability distributions

Normal distribution

$$\phi(x) \equiv f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

Power-transformed distribution

$$\phi(x) \equiv f(x; \mu, \sigma^2, \lambda) = \frac{\lambda x^{\lambda-1}}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x^\lambda-\mu)^2}$$

When $\lambda < 1$, the right side of the graph is longer than than the left. When $\lambda > 1$, the left side is longer than the right. An asymmetry where one function of the graph is longer than the other is called skewness.

An example of a power-transformed distribution is the log distribution,

$$f(x; \mu, \sigma^2) = \frac{1}{x\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(\ln(x)-\mu)^2}$$

more distributions - logistic - duration (exponential, pareto, gamma, weibull)

Week 3

Estimands, estimates, estimators...

A random sample, $\{Y_1, Y_2, ..., Y_n\}$, is a set of random 'variables'. It is a proper subset of the population where each member of the population has an equal probability of being selected. When you calculate a sample statistic, the statistical *function* is applied on the observed sample (the input). This function is called an *estimator* and its output an *estimate*. Formally, an estimator is a function

$$f : \{Y_1, Y_2, ..., Y_n\} \mapsto \mathbb{R}$$

where the estimates are a subset of the set of real values, $\mathbb{R}$.

In a (hypothetical) indefinite repeated-samples scenario, threre will be variation in the mean on account of not every sample containing the exact same members Y_i. An estimate from a sample will not be identical to the poulation value, with the difference being the *sampling error*. Graphing these samples along the values it takes also gives us an indicator of the *sampling variability*.

To formalise our expression of uncertainty about our estimates, we use the standard error. Using prior knowledge of sampling variability on repeated samples as a starting point, we can expect

that our different estimates across different repeated samples yields a probability distribution of all the different estimates, known as the *sampling distribution*. Calculating the spread of this distribution then gives us the *standard error*.

Important: Note that our usage of the term 'variable' here differs from that which is found in lay statistical usage. While Y is some random variable across all members of the population, Y_i is the function localised to each observation i. Thus, the term 'variable' here is more analogous to an observation.

Confidence intervals

A confidence interval can be interpreted as the range in which repeated samples, over the long run, will contain the true mean with some given probability (often 95%).

Important: A 95% confidence interval does not mean that there is a 95% probability for the population parameter to lie inside some *specific* confidence interval, $[y - m, y + m]$. Rather, it pertains to the *a priori generalised CI function* $[Y - 1.96\sqrt{n}, Y + 1.96\sqrt{n}] \quad \forall n$ prior to the value y being specified. That is, before the sample is drawn, we have 95% confidence that the sample we draw contains the population mean μ. Once the value has been decided and the samples taken, the CI either contains the (unknown) population parameter or it doesn't, with no way to tell for sure - though of course, we can be less or more confident about it.

It is also possible to reframe this 'prospecting' confidence in retrospect: if we repeat the procedure indefinite times, the CIs we come up with, regardless of the exact sample estimates, will contain the sample mean 95% (or more generally, $100(1 - \alpha)\%$) of the time. The more samples we take, the more convergent to 95% it gets.

However, this estimation has another pitfall: it assumes that we know the population's standard deviation σ. Oftentimes, we cannot, and what we have is instead the estimate $\bar{s}$. Since we cannot use this, we have to relax our assumption of normality and use the t distribution instead.

Finite sample properties of estimators

Here, we talk about biasedness, variance, and relative efficiency. Both of these pertain to an increasingly large *number of draws*.

Unbiased

An estimator is unbiased when the mean of the estimator distribution, given some fixed sample size, converges to the expected value of the parameter over the course of an indefinite amount of draws

Relative efficiency

(Relative) efficiency is when, given two unbiased estimators W_1 and W_2, the more efficient estimator W_1 has

- Variances smaller than/equal to the other's variances on all values of the parameter
- A strictly smaller variance on at least one value of tne parameter

Formally

$$\forall\theta, \mathrm{Var}(W_1) \leq \mathrm{Var}(W_2) \text{ and } \exists\theta : \mathrm{Var}(W_1) < \mathrm{Var}(W_2)$$

Asymptotic properties of estimators

Asymptotic measures like consistency pertain to an increasingly large *sample size*.

Consistency

Hypothesis testing

We will never know for sure the (*a posteriori*) odds of some sample statistic/estimator at hand producing an erroneous sample estimate. However, we can find out the *a priori* probability that our (interval) estimate may not contain the estimand (population parameter).

Levels α of a statistical significance tests are simply some given probability thresholds of a Type I error occurring. Formally, α is the the null hypothesis being rejected given that it actually happens:

$$\alpha = \Pr(\text{Reject } \ H_0 | H_0)$$

Approaches to assessing uncertainty

Analytic method

This approach presumes a normal distribution

Simulation

This approach also presumes a normal distribution

The following instruction is specific to estimating uncertainty of differences in means.

In programming terms,

- One first creates SE values for each group. Here, it may be a good exercise to see how differences in the number of data points translates to differences in how wide/narrow the SE ranges are.

Bootstrapping

- First, pull all the data points by the grouping variable/s of interest.
- Then, create the actual simulations of sampling by creating a tibble using `expand_grid()`. Here, the number of rows being defined by the number of bootstrap simulations to be conducted, and the number of columns is the sample size.
- Afterwards, group the *tibble* data by row (by each bootstrap simulation).
- Lastly, summarise the mean *including* the argument `replace = TRUE`.

Week 4

Basic intuition

Assume a two-dimensional scatterplot for which we assume a linear relationship between the two variables. The entire set of all the points, (X_i, Y_i), in the scatterplot of dependent variable Y against explanatory variable X has both a set of x-values, x_i, and y-values, y_i. These values

therefore have a mean along both the x-axis, $\bar{X} = \bar{x}$, and the y-axis $\bar{Y} = \bar{y}$. Given that we expect most (if not all!) data points to be offset from $\bar{X}$ and $\bar{Y}$ (and therefore, their respective values offset from $\bar{x}$ and $\bar{y}$)[1], we expect to see a difference between the point values and the mean value, $y_i - \bar{y}$.

Some of this deviation is going to be explained by the line, which gives us the predicted values $\hat{x}, \hat{y} \quad \forall x \in X$, but social science data is (at least overwhelmingly) not perfectly linear. Therefore, we additionally expect residuals, ϵ_i, as the deviations from the mean that are unexplained by the line, i.e., $y_i - \hat{y}$. Given the linear regression formula

$$\hat{y} = \beta_0 + \beta_1 \hat{x}$$

we can re-formulate E_i as

$$\begin{aligned} \epsilon_i &= y_i - \hat{y} \\ &= y_i - (\beta_0 + \beta_1 \hat{x}) \end{aligned}$$

Going forward, it is important to remember that the sum of residuals equals zero. Why is the sum of residuals $\epsilon_i = \sum(y_i - \bar{y}) = 0$, as well as the sum of x-axis deviations from the mean $\sum(x_i - \bar{x}) = 0$?

Let m_i be a series of values $m_1, m_2, ..., m_n$. Let its mean be $\bar{m} = \frac{1}{n}\sum_{i=1}^{n}(m_i)$. Therefore, $m_i = \bar{m} + (m_i - \bar{m})$, the latter of which we can abstract away as $\delta_i = m_i - \bar{m}$. This means that

$$\begin{aligned} \bar{m} &= \frac{1}{n}\sum_{i=1}^{n}(\bar{m} + \delta_i) \\ &= \frac{1}{n}(\sum_{i=1}^{n}\bar{m} + \sum_{i=1}^{n}\delta_i) \\ n\bar{m} &= \sum_{i=1}^{n}\bar{m} + \sum_{i=1}^{n}\delta_i \\ &= n\bar{m} + \sum_{i=1}^{n}\delta_i \\ \sum_{i=1}^{n}\delta_i &= n\bar{m} - n\bar{m} \\ &= 0 \end{aligned}$$

Let either $m_i = x_i$ and $\delta_i = x_i - \bar{x}$ or $m_i = y_i$ and $\delta_i = y_i - \bar{y}$. Therefore

$$\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n}(y_i - \bar{y}) = 0 (4.1)$$

[1]This concerns an ultimately technical distinction, but given the importance of separating concepts from their values, I decided to go through the trouble of writing this out.

This finding motivates us to operationalise 'minimum residuals' differently than simply adding up their values. The standard method of doing this is by squaring them. We do this for three reasons:

1. We remove the cancelling effect of negative-sign residuals (common to techniques like LAV)
2. We penalise outliers more in calculating the line of best fit (specific to power transformation)
3. It is more mathematically tractable, allowing partial derivation to obtain coefficient values

Linear regression

The above minimisation problem can be formulated as

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Given that we have the entire term squared, we can (partially) differentiate to find out the first-order conditions given a change on *either* $\hat{\beta}_0$ or $\hat{\beta}_1$.

$$\begin{aligned}
\frac{\partial}{\partial \hat{\beta}_0} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 &= \sum_{i=1}^{n} 2(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(-1) \text{ by chain rule} \\
&= -2 \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \qquad \stackrel{!}{=} 0 \\
\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) &= 0 \\
\sum_{i=1}^{n} y_i - n\hat{\beta}_0 - \sum_{i=1}^{n} \hat{\beta}_1 x_i &= 0 \\
\sum_{i=1}^{n} y_i - \hat{\beta}_1 \sum_{i=1}^{n} x_i &= n\hat{\beta}_0 \\
\hat{\beta}_0 &= \frac{1}{n} \sum_{i=1}^{n} y_i - \hat{\beta}_1 (\frac{1}{n} \sum_{i=1}^{n} x_i) \\
&= \bar{y} - \hat{\beta}_1 \bar{x} \qquad (4.2)
\end{aligned}$$

Given this formulation of $\hat{\beta}_0$, any OLS estimator thus passes through both the x-axis and y-axis mean values. Now, to find $\hat{\beta}_1$, repeat the FOC differentiation but on $\hat{\beta}_1$

$$\frac{\partial}{\partial \hat{\beta}_1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^{n} 2(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(-x_i) \qquad \text{by chain rule}$$

$$= -2 \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(x_i) \qquad \stackrel{!}{=} 0$$

$$\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(x_i) = 0$$

$$\sum_{i=1}^{n} (y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i)(x_i) = 0 \qquad \text{sub in (4.2)}$$

$$\sum_{i=1}^{n} (y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i)(x_i) = 0$$

$$\sum_{i=1}^{n} (y_i - \bar{y} - \hat{\beta}_1 x_i + \hat{\beta}_1 \bar{x})(x_i) = 0$$

$$\sum_{i=1}^{n} ((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}))(x_i) = 0$$

$$\sum_{i=1}^{n} (y_i - \bar{y})(x_i) - \hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(x_i) = 0$$

$$\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(x_i) = \sum_{i=1}^{n} (y_i - \bar{y})(x_i)$$

$$\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(\bar{x} + x_i - \bar{x}) = \sum_{i=1}^{n} (y_i - \bar{y})(\bar{x} + x_i - \bar{x})$$

$$\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(\bar{x}) + \hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x}) = \sum_{i=1}^{n} (y_i - \bar{y})(\bar{x}) + \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})$$

$$\bar{x}\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x}) + \hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x}) = \bar{x} \sum_{i=1}^{n} (y_i - \bar{y}) + \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})$$

$$\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x}) = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) \qquad \text{sub in (4.1)}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x})} \qquad (4.3)$$

$$= \frac{\text{Cov}(x, y)}{\text{Var}(x)}$$

$$\beta_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

The slope coefficient, $\hat{\beta}_1$ represents the *marginal coefficient*

Goodness of fit

TSS = ESS + RSS

ESS is $(\hat{Y} - \bar{Y})$ and RSS is $(Y_i - \hat{Y})$

Assumptions

A1. Linearity

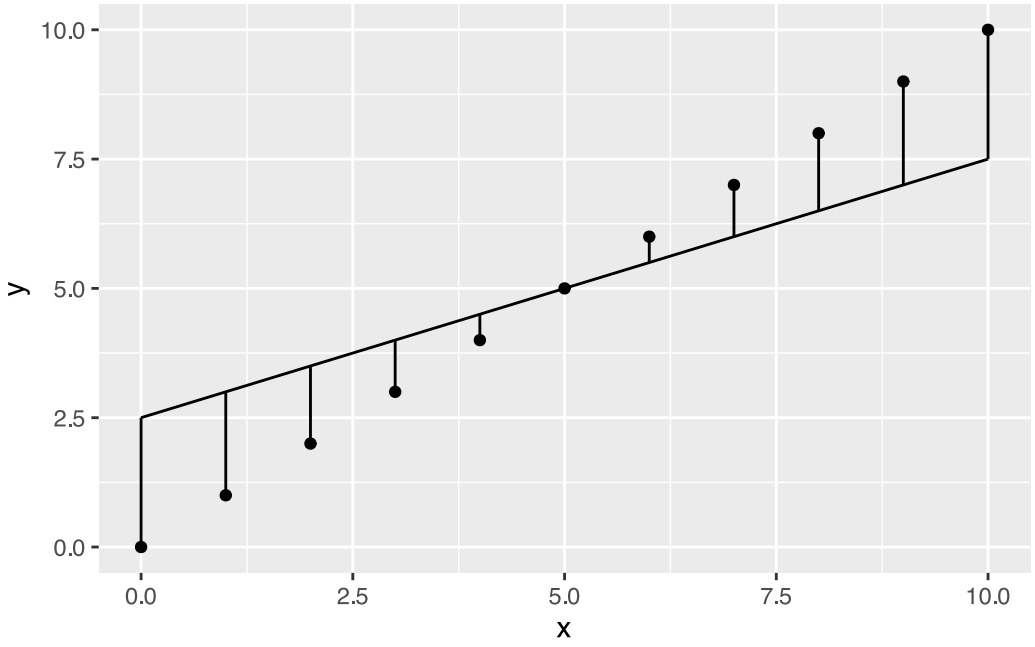
The coefficients should be linear. This is self-explanatory and violation of linearity implies, at minimum, violation of A2.

A2. Zero conditional mean

The expected value of the standard error of i, given some X, is 0:

$$(\epsilon_i | X) = 0$$

This implies that the error terms should not be correlated with the independent variable. We will cover two potential scenarios involving violation of this assumption. The first is when the slope is misidentified (and therefore the marginal effect biased):



Regressing the residuals in the graph above against Y gives us a (statistically significant) correlation, when we expect none or a non-significant one:

	(1)
Intercept	-2.500
X-value	0.500
Num.Obs.	11
R2	1.000

The second is when relationship is not linear (which also means A1 is violated).

A3. Constant variance (homoscedasticity)
The variance should remain constant regardless of X

$$\text{Var}(\epsilon_i|X) = \sigma^2$$

A4. Uncorrelated errors
A violation of the uncorrelated errors assumption means that the estimated standard errors are wider than

Broadly speaking, there are three types of violation here. The first involves positively correlated errors:

A5.
The error terms are distributed normally

Week 5

Week 6

Confidence intervals

What are confidence intervals?
Confidence intervals (CIs) are an expression of uncertainty about some estimate's precision, constructed such that the true population mean will be between an analytically calculated (class of) interval(s) some high percentage of the time, usually 95%. The level of 'error' (that is, the probability that the population mean is outside of this class of ranges) is called the 'significance level', $\alpha \in (0, 1)$.

Now to the construction of the confidence interval itself: let some α and let δ be a (+)ve number. The CI around some marginal effect β_j is defined as

$$\Pr(\hat{\beta}_j - \delta \leq \beta_j \leq \hat{\beta}_j + \delta) = 1 - \alpha \quad (6.1)$$

That is, the probability that the true population mean is between (the estimate ± δ) equals one minus the significance level.

A remark on CI interpretations

Let $\hat{\beta}_j$ have some sign, whether it be $(+)$ or $(-)$. If the CI doesn't include the zero point, we accept some 'possibility of variation' (i.e., imprecision) in the *value* of the actual population parameter, but we are $(1-\alpha)$ confidence *a priori* that the population parameter agrees with our *interval estimate*[2], and perhaps slightly more so in terms of the sign. Ergo, one can say that they qualitatively agree, although to some this might not be sufficient[3].

If a CI does include the zero, however, then it is a sufficient (albeit not *always* necessary[4]) condition for qualitative disagreement. The CI including the zero point means that we have an *a priori* epectation that there is a non-negligible[5] probability of the coefficient/marginal effect being either

- null, or
- having the opposite effect to the one expected.

The two ways to calculating CIs

One way to calculate CIs is to presume known population variance (ergo, also the coefficient SEs) and assume a normal sampling distribution. We thus construct our CIs with $\delta \approx 1.96SE(\beta_j)$. This is also generally applicable whenever we have $n > 120 + k + 1$ observations ($n - k - 1 > 120$ DOF), with k being the number of explanatory variables.

However, we don't always have large enough samples - or perhaps, we have 'too many explanatory variables'. We can relax this assumption of known population variance and instead use a different (non-normal) distribution, the t distribution. This is given as

$$t^* = \frac{\hat{\beta}_j - \beta_j}{\mathrm{SE}(\hat{\beta}_j)} \tag{6.2}$$

$$= \frac{\hat{\beta}_j - \beta_j}{\sqrt{\frac{\hat{\sigma}^2}{\sum_{i=1}^{n}(x_{ji}-\bar{x})^2}}} \sim t_{(n-k-1)}$$

with the SE recalculated to utilise the estimated variance, causing the test statistic to no longer be normally distributed. The t distribution depends on the DOF in such a way that it is similar enough to the normal distribution with a high enough DOF (usually, more observations) for the normal distribution to be usable. We now have the CI

$$\Pr(t_{(\frac{\alpha}{2})} \leq t^* \leq t_{(1-\frac{\alpha}{2})}) = 1 - \alpha \tag{6.3}$$

[2]To reiterate, this means that in the long run, $(1-\alpha)$ of all *repeated trials* will contain the population interval given this CI *specification*.

[3]Whether this is a sufficient criterion for qualitative agreement, or whether the absolute values of the estimates themselves actually matter, is for sound judgment to decide. Even then, sound judgment should still be deployed to determine whether the possible values in the confidence intervals represent wide enough of a 'span' to change its associated substantive conclusions.

[4]See preceeding footnote

[5]Whether some level of significance, often 5%, constitutes a negligible percentage is a well-worn debate topic, though I certainly don't think it can ever be 'too worn'

Recall equation (6.1). In words, we understand this as $1-\alpha$ being the probability that the true population parameter is between estimate $-\ \delta$ and estimate $+\ \delta$. Therefore, the boundary values of the CI signify $\beta_j = \hat{\beta}_j - \delta$ and $\beta_j = \hat{\beta}_j + \delta$ respectively. Therefore, the δ value at the lower bound is

$$\delta = \hat{\beta}_j - \beta_{jl}$$

and the δ at the upper bound is

$$\begin{aligned}\delta &= \beta_{ju} - \hat{\beta}_j \\ &= -(\hat{\beta}_j - \beta_{ju})\end{aligned}$$

and therefore

$$\begin{aligned}\hat{\beta}_j - \beta_{jl} &= -(\hat{\beta}_j - \beta_{ju}) \quad (6.4)\\ (&= \delta \quad)\end{aligned}$$

Revisiting equation (6.3), we verbally understand the equation as the probability that t^* is larger than or equivalent to the t-value for some DOF at $\frac{\alpha}{2}$[6] and smaller than or equal to t at $(1-\frac{\alpha}{2})$.

Revisiting equation (6.2) and bringing (6.4) gives us $\delta = t^*\ \mathrm{SE}(\hat{\beta}_j)$, where the lower boundary is

$$\beta_{jl} = \hat{\beta}_j + t_{(\frac{\alpha}{2})}\ \mathrm{SE}(\hat{\beta}_j)$$

and the upper boundary

$$\beta_{ju} = \hat{\beta}_j + t_{(1-\frac{\alpha}{2})}\ \mathrm{SE}(\hat{\beta}_j)$$

Given that $\beta_{ju} = \hat{\beta}_j + \delta = \hat{\beta}_j + t^*\ \mathrm{SE}(\hat{\beta}_j)$ corresponds to the upper boundary, we can infer that $t^* = t_{(1-\frac{\alpha}{2})}$. By symmetry,

$$\begin{aligned}t_{(\frac{\alpha}{2})}\ \mathrm{SE}(\hat{\beta}_j) &= -\delta \\ \hat{\beta}_j + t_{(1-\frac{\alpha}{2})}\ \mathrm{SE}(\hat{\beta}_j) &= \hat{\beta}_j - t_{(\frac{\alpha}{2})}\ \mathrm{SE}(\hat{\beta}_j)\end{aligned}$$

and therefore

$$\Pr(\hat{\beta}_j - t_{(1-\frac{\alpha}{2})}\ \mathrm{SE}(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + t_{(1-\frac{\alpha}{2})}\ \mathrm{SE}(\hat{\beta}_j)) = 1-\alpha \quad (6.5)$$

Errors, errors

Statistical inference essentially boils down to deciding between two alternative (substantive) interpretations: either there is no ('null') effect, H_0, or there is an effect, H_A. The latter is the 'alternative' hypothesis, hence the A subscript.

[6] two-tailed, therefore the probability of the true population param lying outside the CI, which is what α essentially is, is divided equally for each of the two tails

With a 95% CI (I hope what this means is clear enough to the reader by now), H_0 being true implies that we falsely reject H_0 ('false positive') 5% of the time *in the long run*. That is, $\Pr(\text{reject } H_0 | H_0) = 0.05$.

The α-bsolutist way...

In testing the statistical significance of a regression coefficient, we want to determine whether our estimated coefficient is non-null/non-zero, i.e., $\hat{\beta}_j \neq 0$. Therefore, we use a two-tailed hypothesis to test against H_0:

- H_0: $\beta_j^* = 0$, where $*$ denotes 'test'[7]
- H_A: $\beta_j^* \neq 0$

Thus, we construct t^* given our hypotheses about $\hat{\beta}_j$. Bringing back equation (6.2) and given a baseline expectation of null (i.e., $\beta_j^* = 0$),

$$\begin{aligned} t^* &= \frac{\hat{\beta}_j - \beta_j^*}{\text{SE}(\hat{\beta}_j)} \\ &= \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \sim t_{(n-k-1)} \end{aligned}$$

We reject H_0 at the α level if $|t^*| > t_{(1-\frac{\alpha}{2}[,n-k-1])}$.

...and the p-retty nuanced way

So far, we've framed statistical significance in binary terms, i.e., whether our test statistic clears some 'arbitrary' hurdle. What if different people and different venues set different hurdles? The concept of p-values reframes the expression of uncertainty and states it as the 'maximum height of the hurdle' which our test statistic can clear.

To return to our α example, our hurdle is $t^* = t_{(1-\frac{\alpha}{2},n-k-1)}$. When it is cleared, we are $100(1-\alpha)$ confident of a CI. So, what about the reverse? The probability that we misdiagnosed the situation and $|t^*|$ is actually smaller than $|t_{(n-k-1)}|$[8]?

Once again, $\Pr(t^* \geq t_{(1-\frac{\alpha}{2},n-k-1)}) = 1 - \alpha$. Generalised,

$$\Pr(t_{(n-k-1)} \leq t^*) = 1 - p$$

The p-value therefore takes the exact opposite scenario; neither $>$ nor $=$, but just $>$:

$$\Pr(t_{(n-k-1)} > t^*) = p$$

$\Pr(t > t^*)$ denotes the area to the right of t^* given $n-k-1$ DOF. By symmetry,

$$\begin{aligned} p &= \Pr(|t_{(n-k-1)}| > |t^*|) \\ &= 2\Pr(t_{(n-k-1)} > |t^*|) \end{aligned}$$

[7] See t^* in the previous section

[8] Since we're not dealing with αs here, $\frac{\alpha}{2}$ is irrelevant

The above gives us the smallest significance level at which H_0 can be rejected, given the resulting $t^*_{(n-k-1)}$.

Categorical/binary variables

'Qualitative' (nominal, sometimes ordinal) explanatory variables[9] can be reduced to one, or a series of, binary (often called 'dummy') variables. When reducing/recoding to binary variables, it is good practice to

- Code the two categories within each binary variable as 0 and 1,
- Assign the 0 value to some baseline, and
- Name variables after the non-baseline ('1') category.

Evading the binary variable trap

Let's take a polytomous variable, such as 'party an individual voted for', as an example. If each category gets its own binary variable, the resulting binary variables will be perfectly collinear as they all take the value 1. Given k categories, we avoid this collinearity trap by splitting this into $k-1$ binary variables, with the baseline category taking on the 0 value in all binary variables. Note that coefficient interpretation is always relative to the baseline category.

For polytomous nominal variables, simply assign one binary variable for non-baseline category with a 1 for the *respective* category it identifies and zero for all the others. For ordinal variables, it may help to keep the 1 instead of turning it to a 0 as we go up the scale. This allows us to calculate the coefficients between levels with standard error calculation included.

Interpreting interaction coefficients

Let's use the following example:

$$
\begin{aligned}
Y &= \beta_0 + \beta_1 X_1 + \beta_2 D + \beta_3 X_1 D + \epsilon \\
\frac{\partial Y}{\partial X_1} &= \beta_1 + \beta_3 D \\
\frac{\partial Y}{\partial D} &= \beta_2 + \beta_3 X_1
\end{aligned}
$$

Focusing on X_1,

- When $D = 0$, the marginal effect of X_1 is β_1.
- When $D = 1$, the marginal effect of X_1 is $\beta_1 + \beta_3$.

Focusing on the intercept,

- When $D = 0$, the intercept is $beta_0$. Quite straightforward so far, however...
- When $D = 1$, $\beta_2 D = \beta_2$ is also a constant, and the intercept is therefore $beta_0 + \beta_2$

We notice two things:

- The marginal effect of X_1 changes in value with different values of D

[9] Regressing them as DVs is the topic of a future section

- The marginal effect of D changes the value of the intercept

This constitutes the ground upon which we explore the rules to follow in interpreting interaction coefficients, as laid out by Thomas Brambor et al. (2007).

Include interaction terms

It should be clear why: doing otherwise obscures the interaction between both variables. This This is a rather trivial point, but it bears repeating because it is perhaps fundamentally connected to the error in interpretation often seen in political science literature which came out more than two decades ago (two subheadings down).

Include all constituent terms

A common mistake in conducting interaction regressions is to exclude e.g., the binary constituent of an interaction effect. Specifically in this context, not specifying all constituent terms forces the regression lines to share an intercept, yielding biased slope coefficients.

Do not interpret marginal coefficients as unconditional

A common flawed practice in social science is to simply report one IV's marginal effects as its own (independent) beta-coefficient. In what follows, I mathematically elaborate why this is incorrect. Suppose an equation with two independent variables, X and Z, which do not interact:

$$\begin{aligned} Y &= \omega_0 + \omega_1 X + \omega_2 Z + \mu \\ \frac{\partial Y}{\partial X} &= \omega_1 \end{aligned}$$

When X and Z do not interact, reporting beta coefficients independently is not a problem. Now let's assume they do and add an interaction term:

$$\begin{aligned} Y &= \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + \epsilon \\ \frac{\partial Y}{\partial X} &= \beta_1 + \beta_3 Z \\ \frac{\partial Y}{\partial Z} &= \beta_2 + \beta_3 X \end{aligned}$$

This highlights that the marginal effect of either IV is not its beta-coefficient alone.

Calculate substantively meaningful marginal effects and SEs

Oftentimes you'll only be dealing with a binary interaction variable. Assuming such a case involving two interaction variables, and given that you're calculating marginal effects primarily from the perspective of the continuous variable, it's a straightforward task to interpret the marginal coefficients:

$$\begin{aligned} & Y = \beta_0 + \beta_1 X_1 + \beta_2 D + \beta_3 X_1 D + \epsilon \\ (D = 0) \quad & Y = \beta_0 + \beta_1 X_1 + \epsilon \\ (D = 1) \quad & Y = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) X_1 + \epsilon \end{aligned}$$

Theoretically, there are only two sets of values (and four values total, the intercepts and slopes when $D = 0$ and $D = 1$, respectively) you'd need to report.

Interpreting continuous interactions is trickier, but there are simple tricks (e.g., using extreme values of the interaction variable or plotting a graph of how the coefficient changes with the interaction variable). For a good example of this, see Golder (2005).

Additionally, interaction estimates should be accompanied by the correct standard error calculation formula. The standard error of X_1's marginal effect on Y, $\frac{\partial Y}{\partial X_1}$ is

$$\hat{\sigma}_{\frac{\partial Y}{\partial X_1}} = \sqrt{\text{Var}(\hat{\beta}_1) + X_2^2 Var(\hat{\beta}_3) + 2X_2 \ \text{Cov}(\hat{\beta}_1, \hat{\beta}_3)}$$